

Static Pods in Kubernetes

Since you're learning **KCNA as a beginner**, let's understand Static Pods step by step.

What is a Static Pod?

A Static Pod is a Pod that is managed directly by the Kubelet on a Node, instead of being managed by the Kubernetes API Server.

Easy memory:

 **Kubelet = directly manages Static Pods**

Normal Pod vs Static Pod

Normally, Kubernetes works like this:

User

|



API Server

|



Scheduler

|



Kubelet

|



Pod

But with a **Static Pod**:

Static Pod YAML

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Kubelet



Pod

The **Kubelet directly watches a specific directory** for Static Pod definitions.

Real-Life Example 🏠

Imagine a manager normally tells a worker:

"Build this room."

The manager is like the **API Server**.

But for a Static Pod, the worker has a file sitting directly on their desk:

 pod.yaml

The worker sees it and says:

"I need to create this."

The worker is the **Kubelet**.

So:

Static Pod = Kubelet manages it directly.

Where are Static Pods Defined?

The Kubelet is configured with a **static Pod manifest directory**.

A common location is:

/etc/kubernetes/manifests/

For example:

/etc/kubernetes/manifests/



|— pod1.yaml

|— pod2.yaml

└─ pod3.yaml

The Kubelet watches this directory.

When you put a valid Pod YAML file there:

pod.yaml

|



Kubelet detects it

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Kubelet creates Pod

Important Point ★

Static Pods are **not created using**:

kubectl create

or:

kubectl apply

Instead, you create/edit the **Pod manifest file**, and the Kubelet manages it.

Does a Static Pod Appear in `kubectl get pods`?

Yes. This is an important point.

You can generally see a Static Pod through the API Server as a **mirror Pod**.

For example:

`kubectl get pods -A`

You may see a Pod with a name associated with its Node.

But remember:

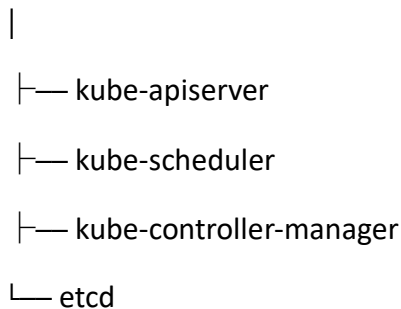
The API Server is **not managing the actual Static Pod**. The Kubelet is.

Why Are Static Pods Used?

Static Pods are commonly used for **important cluster components**, especially control-plane components in some Kubernetes setups.

For example:

Control Plane Node



In setups such as those created by **kubeadm**, these control-plane components can be run as Static Pods.

Static Pod vs Normal Pod

Normal Pod

Managed through Kubernetes control plane

Usually created through API Server

Scheduler normally decides placement

Controlled through Kubernetes objects

Can be created with `kubectl apply`

Static Pod

Managed directly by Kubelet

Created from a local manifest

Runs on the Node where Kubelet manages it

Controlled by the Kubelet

Created from Static Pod manifest

Static Pod vs DaemonSet

You just learned **DaemonSets**, so this comparison is useful.

DaemonSet

DaemonSet

|



Kubernetes Control Plane

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Pods on Nodes

Static Pod

Pod Manifest

|



Kubelet

|



Pod

Main Difference

DaemonSet is managed by Kubernetes. Static Pod is directly managed by the Kubelet.

What Happens If the Kubelet Restarts?

The Kubelet continues watching the Static Pod manifest directory.

If the Static Pod should exist, Kubelet ensures that it is running.

That's one reason Static Pods are useful for important node-level components.

Easy Diagram

STATIC POD

Node

|

|

/etc/kubernetes/

manifests/

|



pod.yaml

|



Kubelet 

|



Pod 

Notice:

API Server ❌

Scheduler ❌

The **Kubelet directly manages the Static Pod.**

Important KCNA Points

Remember these:

- **Static Pod = Pod managed directly by Kubelet.**
- Kubelet gets its definition from a **manifest file**.
- A common manifest directory is:

/etc/kubernetes/manifests/

- Static Pods are tied to the **Node running the Kubelet**.
 - They are commonly used for important node/control-plane components.
 - They can appear through the API Server as **mirror Pods**.
 - The API Server does **not** manage the actual Static Pod.
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Easy Memory Trick

Normal Pod

API Server → Scheduler → Kubelet → Pod

Static Pod

YAML File → Kubelet → Pod

One-line KCNA definition:

A Static Pod is a Pod managed directly by the Kubelet from a local manifest file, rather than being managed by the Kubernetes API Server.